

PolicyGPT: Government Policy & Public

Scheme Intelligence Platform

Description

PolicyGPT is an intelligent web-based platform designed to simplify access to and understanding of government policies and related public information. The platform provides users with a centralized interface through which they can explore policy-related information in a clear, organized, and user-friendly manner.

PolicyGPT aims to reduce the difficulty of finding relevant policy information by providing an easy-to-use digital platform for accessing, understanding, and interacting with policy-related content. The system is designed to improve accessibility and help users obtain relevant information without having to navigate through multiple sources.

The platform provides a structured interface for presenting policy information and supports users in discovering information relevant to their requirements. It combines a modern web interface with organized policy data to provide a simple and efficient user experience.

Key features of PolicyGPT include policy information management, user-friendly policy discovery, personalized access to relevant information, notifications, and administrative functionality for managing platform content.

Overall, PolicyGPT is intended to provide a centralized and accessible platform that makes government policy information easier to discover, understand, and use.

Purpose

The purpose of **PolicyGPT** is to simplify access to government policies, public welfare schemes, regulations, notifications, and other official information through a centralized digital platform. The platform is designed to help citizens and other users quickly search, understand, and access relevant government information.

The main purposes of PolicyGPT are to:

- **Centralize Information:** Provide government policies and schemes through a single, organized platform.
- **Simplify Policy Discovery:** Enable users to search and filter policies based on relevant categories and requirements.
- **Improve Scheme Awareness:** Help citizens discover government schemes and understand their benefits and eligibility.
- **Support Eligibility Checking:** Assist users in identifying schemes for which they may be eligible.

- **Enable Application Tracking:** Allow citizens to apply for schemes and track their application status.
- **Improve Transparency:** Provide government officials with tools to manage policies, schemes, and citizen applications.
- **Provide Timely Updates:** Keep users informed through notifications about policies, schemes, deadlines, and applications.

Overall, PolicyGPT aims to make government information **easier to discover, understand, and access**, while improving transparency and efficiency in public service delivery. These objectives align with the project reference, which emphasizes centralized policy access, intelligent search, eligibility checking, scheme comparison, notifications, dashboards, and analytics.

Getting Started :

Prerequisites

Before setting up and running **PolicyGPT**, ensure that the following software and tools are installed:

- **Node.js and npm** – Required for running the frontend and backend applications.
- **MongoDB / MongoDB Atlas** – Used for storing application data.
- **Visual Studio Code** – Recommended for development and debugging.
- **Git** – Required for version control and accessing the project repository.
- **Modern Web Browser** – Required to access and test the web application.

These prerequisites follow the structure used in the reference documentation.

Installation

1. Clone the Repository

```
git clone https://github.com/springboardmentor212/GPPSIP_Team1.git
```

```
cd GPPSIP_Team1
```

2. Install Backend Dependencies

```
cd backend
```

```
npm install
```

3. Install Frontend Dependencies

```
cd ../frontend
```

```
npm install
```

4. Configure Environment Variables

Create the required .env files and configure the MongoDB connection string and other required environment variables.

5. Run the Application

Start the backend and frontend servers using their respective npm commands.

Running the Application

After completing the environment setup, start the backend and frontend servers.

Start Backend

```
cd backend
```

```
npm install
```

```
npm run dev
```

Start Frontend

Open a new terminal:

```
cd frontend
```

```
npm install
```

```
npm run dev
```

Access the Application

Open the frontend URL displayed in the terminal, typically:

```
http://localhost:5173
```

The backend server will run on the port configured in the environment variables.

Environment Setup

Create a .env file inside the backend directory and add the required environment variables:

```
PORT= 3000
```

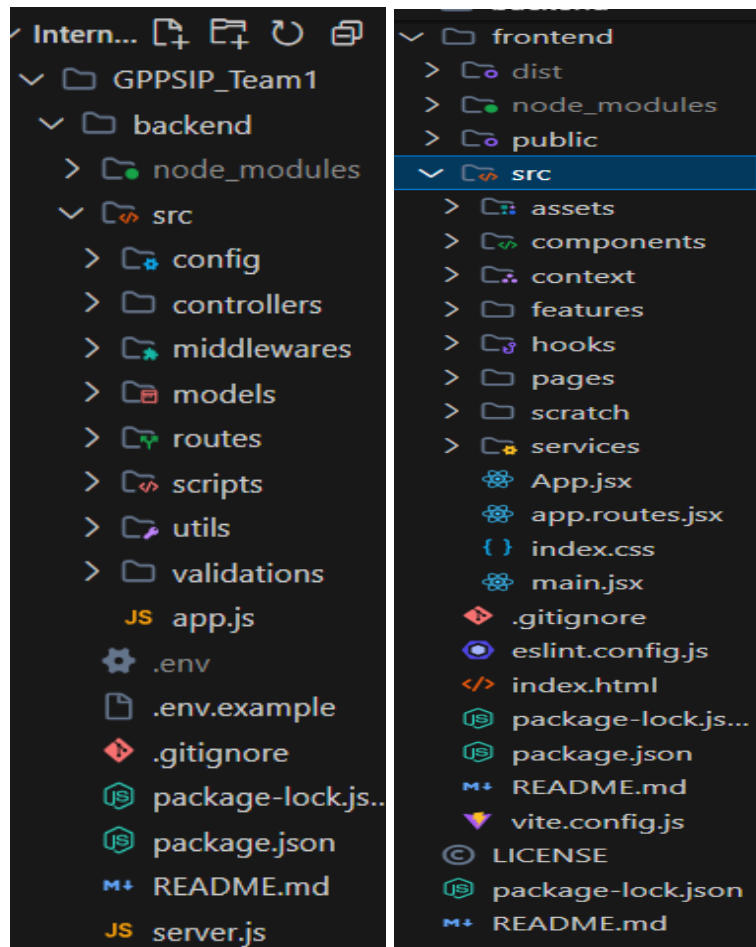
```
MONGODB_URI= mongodb:// 127.0.0.1:27017/policygpt
```

```
JWT_SECRET= policygpt_fallback_secret_key_987654
```

Variable Name	Description	Example Value
PORT	Port on which the backend server runs.	3000
MONGO_URI	MongoDB connection string for the PolicyGPT database.	mongodb://127.0.0.1:27017/policygpt
JWT_SECRET	Secret key used for JWT token generation and verification.	policygpt_fallback_secret_key_987654
NODE_ENV	Specifies the runtime environment of the application.	development

Folder Structure :

The PolicyGPT project follows a modular full-stack architecture with separate frontend and backend directories.



The frontend manages the user interface, dashboards, and API integration, while the backend handles business logic, authentication, database operations, and REST APIs.

Key Features :

PolicyGPT provides the following key features:

- Role-Based Access: Separate access for Citizens, Government Officials, and Super Admins.
- Policy and Scheme Management: View, search, manage, and access government policies and schemes.
- Eligibility Checker: Helps citizens identify schemes based on their eligibility criteria.
- Application Tracking: Allows citizens to apply for schemes and track their application status.
- Approval Workflow: Government Officials can review and approve or reject applications.
- Notifications and Feedback: Keeps users updated and enables feedback submission.
- Administrative Dashboard: Super Admins can manage users, officials, and platform-level activities.

Project Components:

PolicyGPT consists of multiple interconnected components designed to provide a centralized platform for citizens, government officials, and administrators to access, manage, and analyze government policies and schemes.

1. Landing Page Component

Description:

The Landing Page serves as the primary entry point of the PolicyGPT platform and introduces users to its purpose and major services. It provides clear navigation to authentication and other important sections of the platform.

Key Features:

- Attractive interface introducing the PolicyGPT Intelligence Platform.
- Navigation to Login and Registration pages.
- Highlights important platform features and services.
- Provides a user-friendly entry point for different user roles.



1200+

Government Schemes

35+

States & UTs

50+

Ministries

1M+

Citizens Assisted

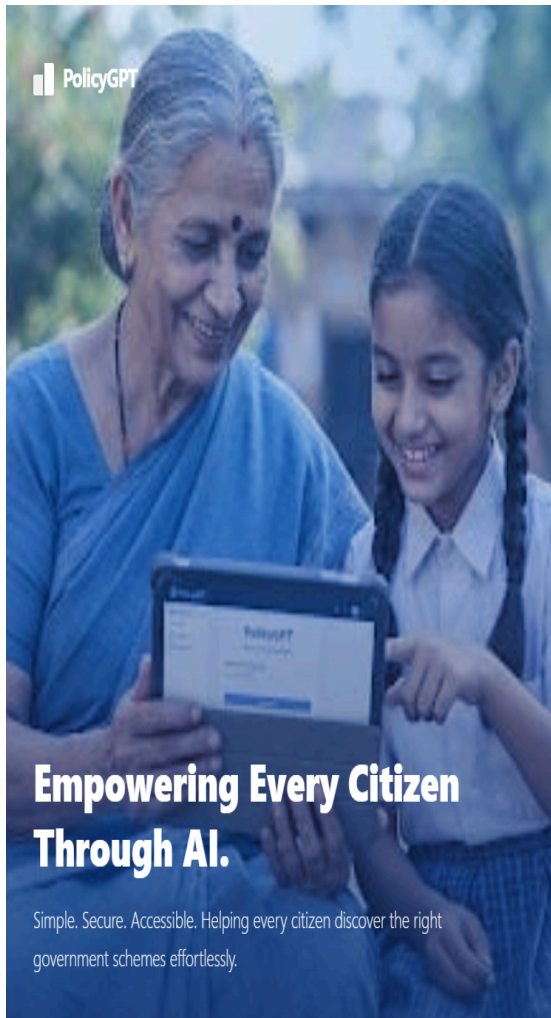
2. Registration Component

Description:

The Registration Component enables new users to create an account on the PolicyGPT platform by providing the required personal and account information. It supports role-based registration to assign appropriate access and functionality.

Key Features:

- User registration with personal and authentication details.
- Role selection for Citizen, Government Official, and Super Admin.
- Form validation for required user information.
- Secure storage of registered user data in MongoDB.



Create Your Account

Join PolicyGPT to receive personalized government scheme recommendations and AI-powered assistance.

FULL NAME	EMAIL ADDRESS
Enter full name	name@example.com
MOBILE NUMBER	DATE OF BIRTH
Enter 10-digit number	dd-mm-yyyy
PASSWORD	CONFIRM PASSWORD
Enter password	Re-enter password
STATE	DISTRICT
Select State	Select District
SELECT YOUR ROLE	
☒ Citizen	☐ Official
☐ NGO	☐ Admin

☐ By creating an account, I agree to the [Terms of Service](#) and [Privacy Policy](#) of PolicyGPT.

[Create Account](#)

Already have an account? [Sign In](#)

Government Official? [Official Verification Form](#)

Researcher / NGO? [Organization Verification Form](#)

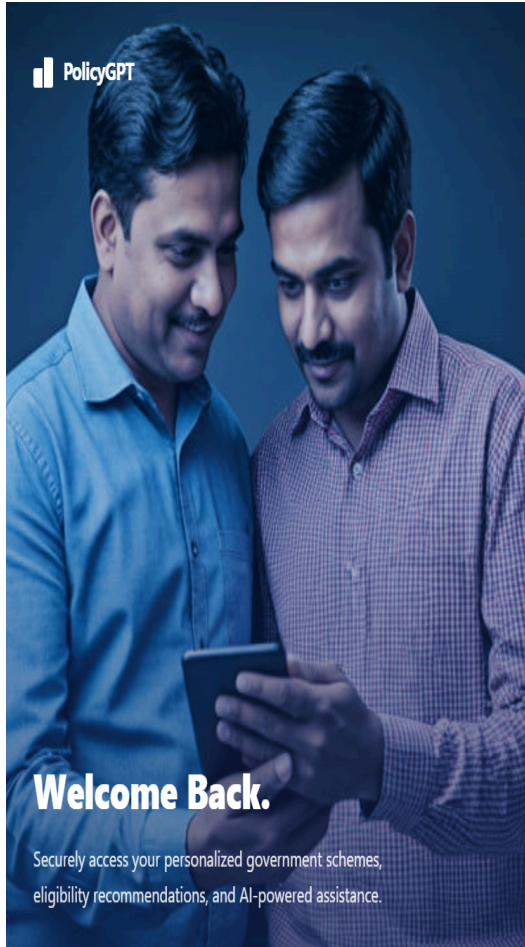
3. Login Component

Description:

The Login Component allows registered users to securely access the PolicyGPT platform using their credentials. Users are authenticated and redirected to the appropriate dashboard according to their assigned role.

Key Features:

- Secure authentication using registered credentials.
- Role-based login and dashboard redirection.
- JWT-based authentication and session handling.
- Error handling for invalid login credentials.



Welcome Back

Sign in to continue exploring government schemes.

EMAIL ADDRESS

PASSWORD

☐ Remember Me [Forgot Password?](#)

[Sign In](#)

Don't have an account? [Create Account](#)

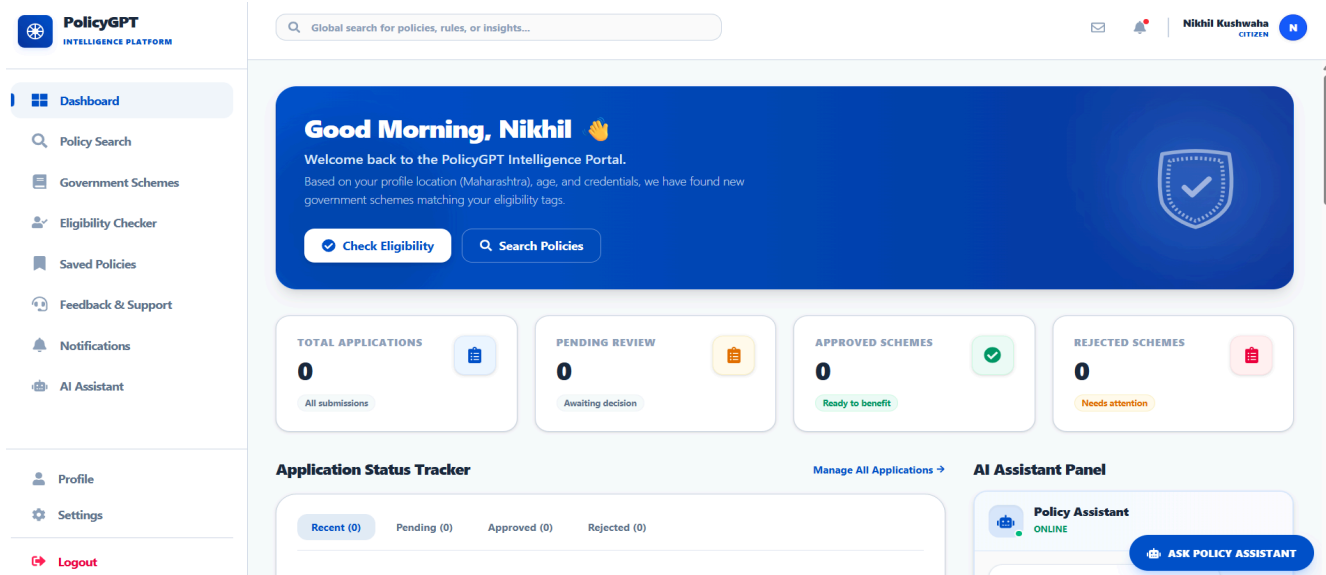
4. Citizen Dashboard Component

Description:

The Citizen Dashboard provides users with a centralized overview of relevant schemes, policies, applications, and platform activities. It acts as the main workspace after a citizen successfully logs into the system.

Key Features:

- Overview of application and scheme-related statistics.
- Quick access to eligibility checking and policy search.
- Application status tracking interface.
- Access to notifications and AI assistance.



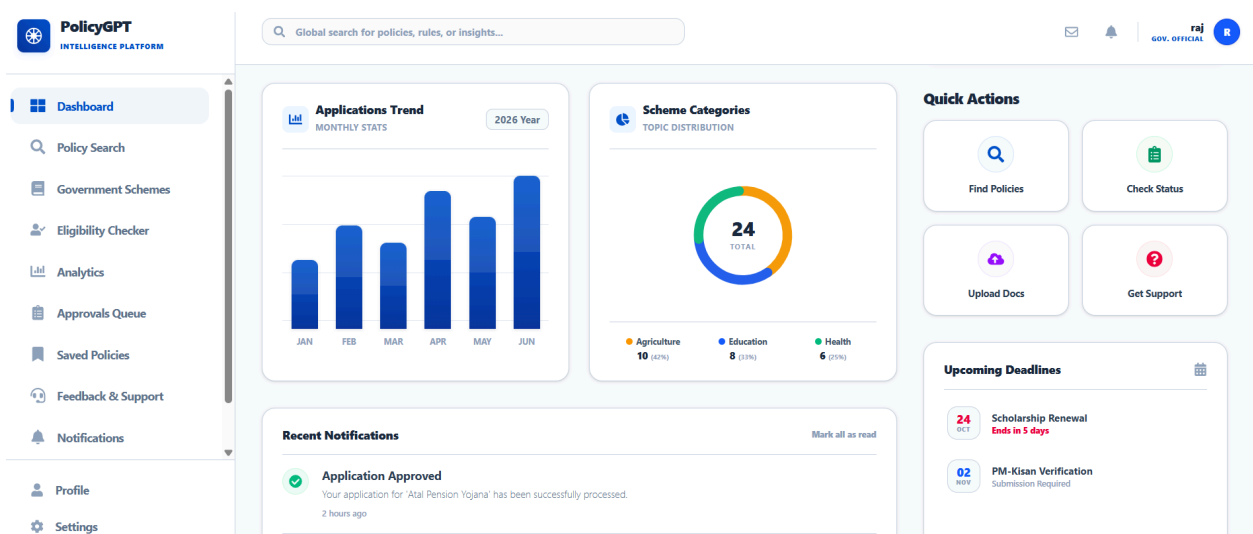
5. Government Official Dashboard Component

Description:

The Government Official Dashboard provides authorized officials with tools to manage citizen applications and related government scheme activities. It helps officials efficiently review and process submitted applications.

Key Features:

- Overview of submitted and pending applications.
- Application review and management functionality.
- Approval and rejection workflow support.
- Access to relevant administrative information.



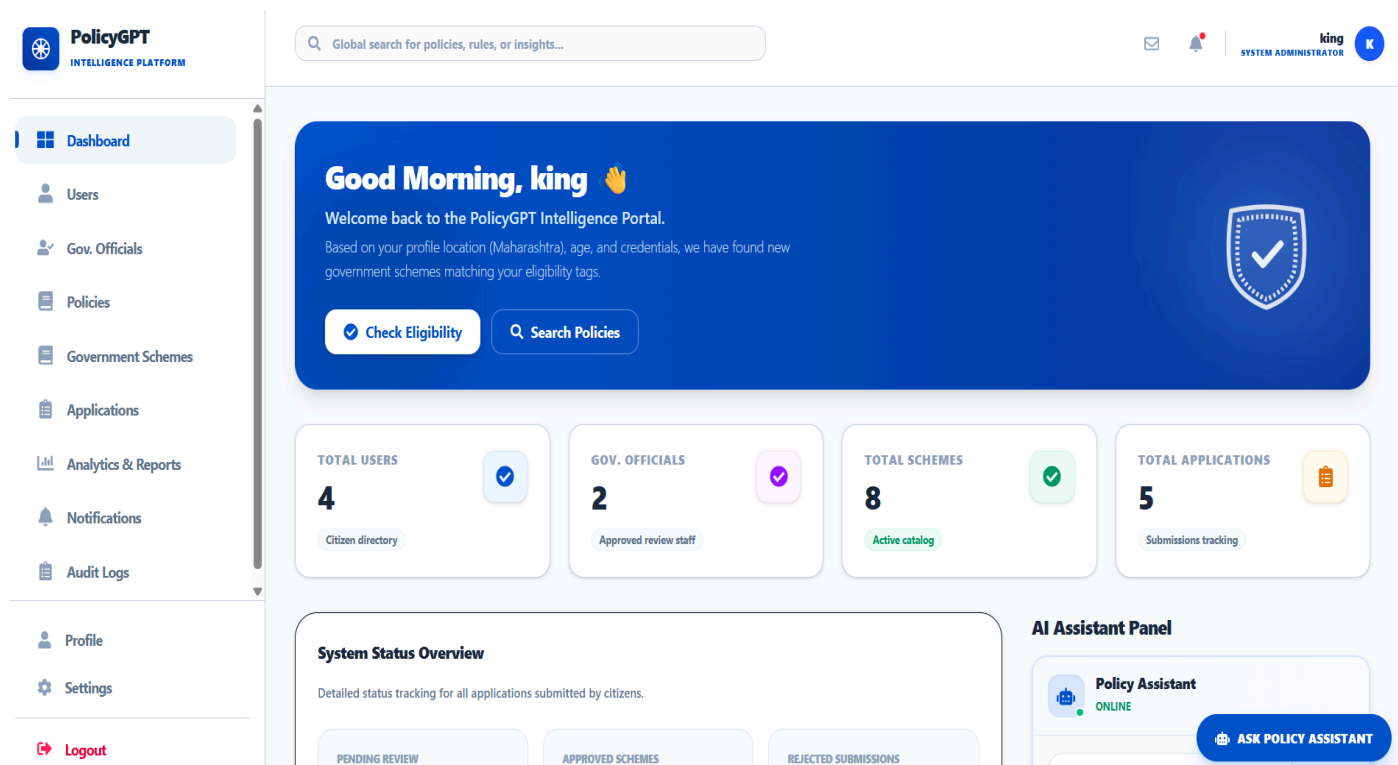
6. Super Admin Dashboard Component

Description:

The Super Admin Dashboard provides centralized administrative control over the PolicyGPT platform. It enables the Super Admin to monitor users, manage platform activities, and access role-specific management features.

Key Features:

- Centralized platform administration interface.
- Management of users and system-level activities.
- Role-specific access for Super Admin accounts.
- Overview of important platform information and operations.



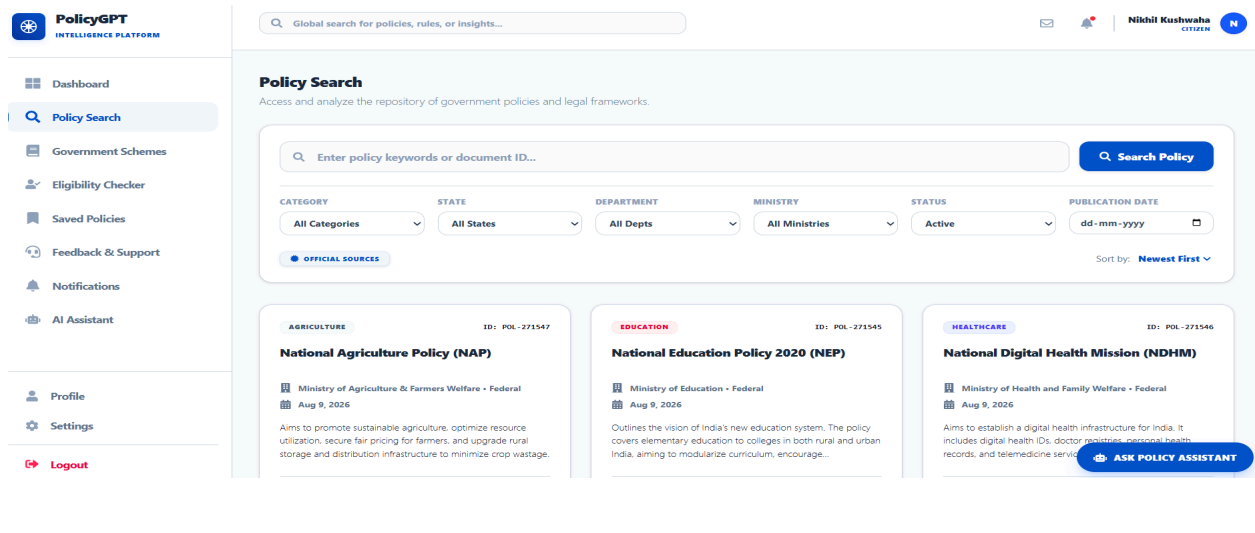
7. Policy Search Component

Description:

The Policy Search Component allows users to search and explore government policies available on the PolicyGPT platform. It helps users quickly locate relevant information according to their requirements.

Key Features:

- Search functionality for government policies.
- Easy navigation to policy information.
- Display of relevant policy details.
- User-friendly search interface.



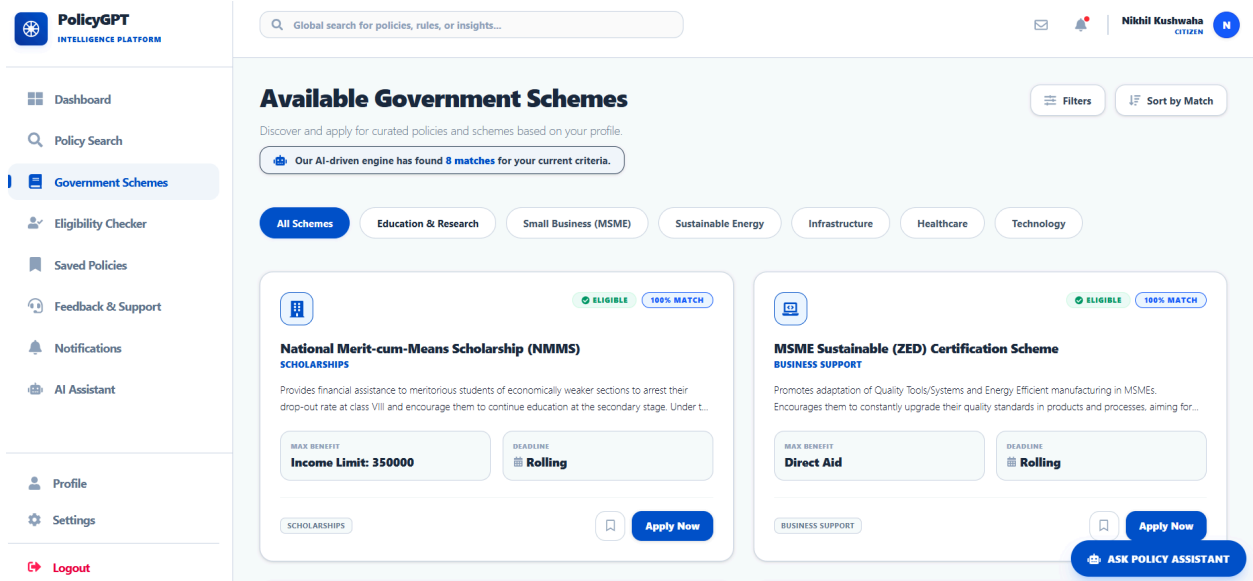
8. Government Schemes Component

Description:

The Government Schemes Component displays available government schemes and allows users to explore their details. Citizens can identify suitable schemes and proceed with the application process.

Key Features:

- Display of available government schemes.
- Category-based scheme exploration.
- Detailed information for individual schemes.
- Direct scheme application functionality.



9. Policy Details Component

Description:

The Policy Details Component provides detailed information about individual government policies selected by the user. It allows users to understand policy objectives, provisions, and other available information.

Key Features:

- Detailed display of policy information.
- Organized presentation of policy content.
- Easy navigation from search results.
- Support for exploring relevant policies.

10. Eligibility Checker Component

Description:

The Eligibility Checker Component helps citizens determine whether they are eligible for a particular government scheme. It evaluates user-provided information against the eligibility requirements associated with schemes.

Key Features:

- User profile-based eligibility checking.
- Matching of user details with scheme requirements.
- Identification of suitable schemes.
- Simplified decision-making before application.

PolicyGPT
INTELLIGENCE PLATFORM

Global search for policies, rules, or insights...

Nikhil Kushwaha
CITIZEN

1 Basic Information 2 Recommended Schemes

Eligibility Assessment

Select a scheme and provide your details to verify eligibility.

TARGET SCHEME *

-- Select a Government Scheme --

AGE: Enter your age

GENDER: Select Gender

ANNUAL INCOME (₹): Enter your annual income in ₹

OCCUPATION: Select Occupation

EDUCATION: Select Education

STATE: Select State

DISTRICT: Select State First

SOCIAL CATEGORY: Select Category

DISABILITY STATUS: Select Option

ASK POLICY ASSISTANT

11. Application Tracking Component

Description:

The Application Tracking Component enables citizens to monitor the status of their submitted scheme applications. It provides visibility into the progress of applications from submission to final decision.

Key Features:

- Tracking of submitted applications.
- Pending, Approved, and Rejected status categories.
- Centralized application status overview.
- Easy access to application-related information.

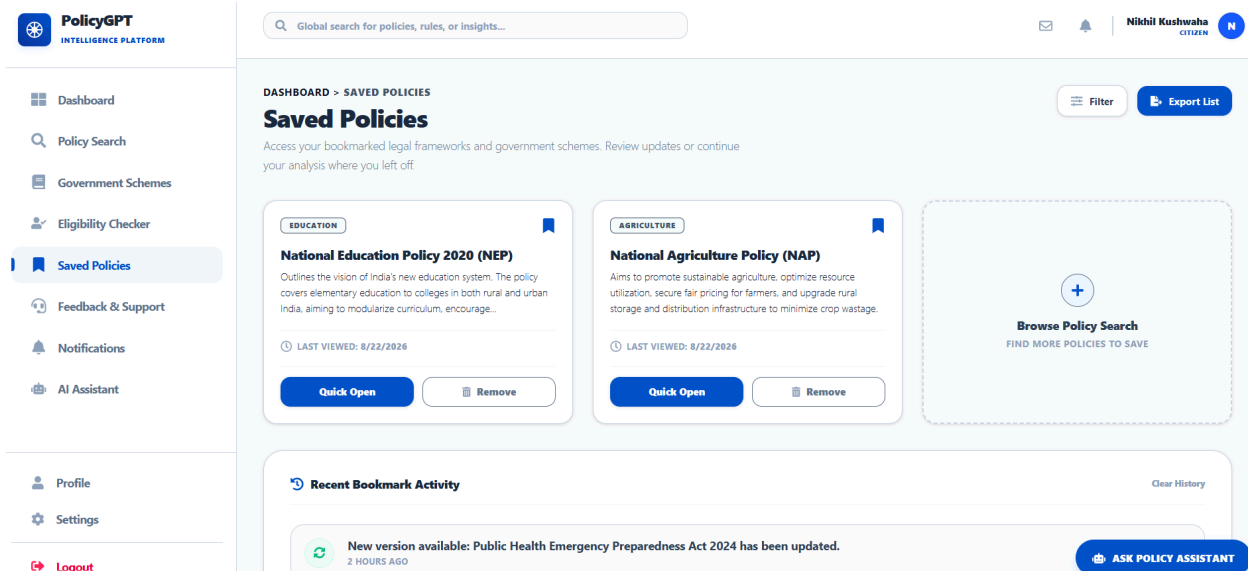
12. Saved Policies Component

Description:

The Saved Policies Component allows users to keep track of policies that they want to revisit later. It provides a personalized area for managing previously saved policy information.

Key Features:

- Save relevant policies for future reference.
- Personalized list of saved policies.
- Easy access to bookmarked information.
- Simplified management of saved content.



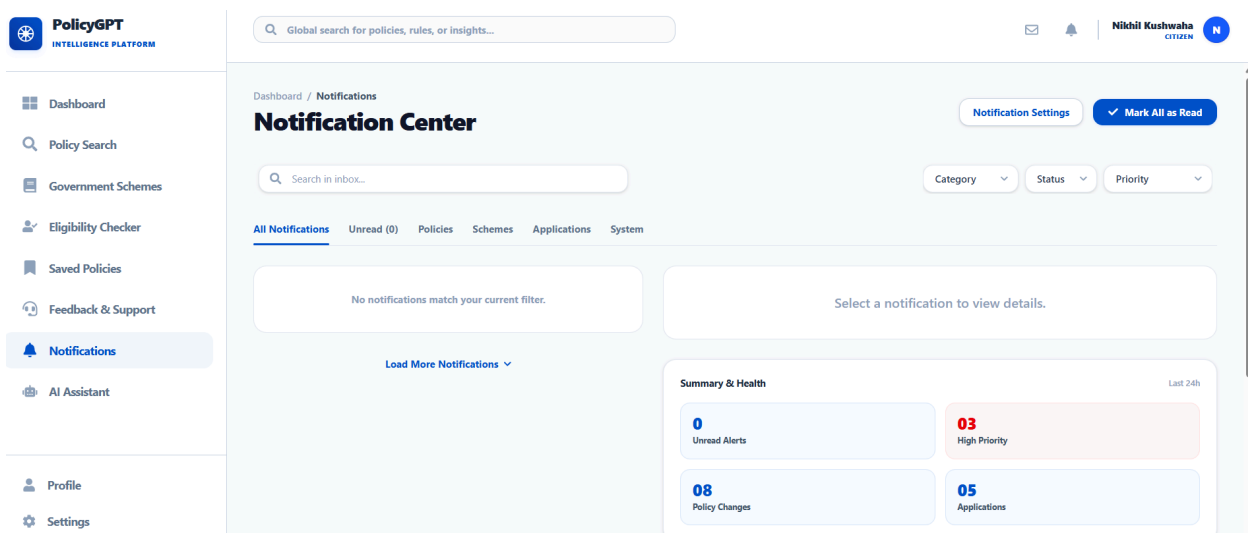
13. Notifications Component

Description:

The Notifications Component keeps users informed about important platform activities and updates. It provides timely information related to applications, approvals, and other relevant actions.

Key Features:

- Centralized notification display.
- Updates related to application activities.
- Important platform alerts and information.
- Role-specific notification access.



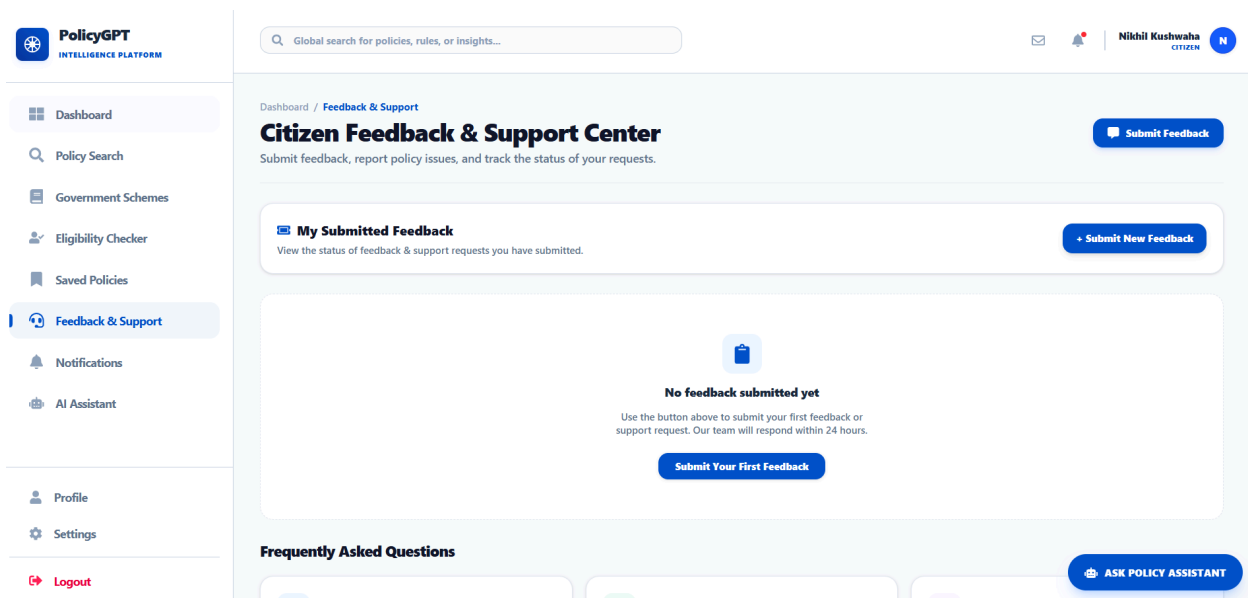
14. Feedback and Support Component

Description:

The Feedback and Support Component enables users to submit feedback, suggestions, or issues related to the PolicyGPT platform. It provides a communication channel between users and the platform administration.

Key Features:

- User feedback submission.
- Support for suggestions and issue reporting.
- Structured feedback form.
- Improved communication between users and administrators.



15. Profile Component

Description:

The Profile Component allows authenticated users to view and manage their personal account information. It provides a centralized location for maintaining user-specific details.

Key Features:

- Display of user profile information.
- Management of account-related details.
- Role-specific user information.
- Personalized account access.

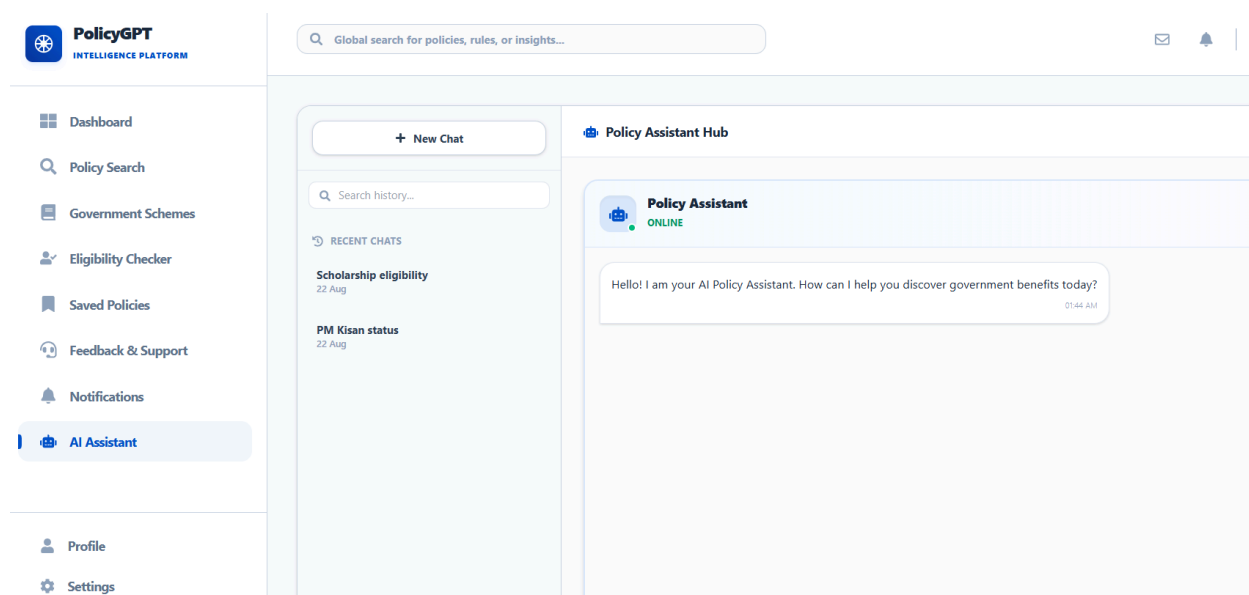
16. AI Policy Assistant Component

Description:

The AI Policy Assistant Component provides an interactive interface through which users can ask questions related to policies, schemes, and platform information. It aims to make government-related information easier to understand and access.

Key Features:

- Interactive question-and-answer interface.
- Assistance with policy and scheme-related queries.
- Easy access to relevant information.
- Conversational and user-friendly interaction.



17. Backend API and Database Component

Description:

The Backend and Database Component manages server-side operations, API requests, authentication, and persistent data storage for the PolicyGPT platform. It connects the frontend interface with the application's business logic and database.

Key Features:

- RESTful APIs built using Node.js and Express.js.
- MongoDB database integration using Mongoose.
- Secure authentication and authorization handling.
- Management of users, policies, schemes, and applications.

Error Handling and Troubleshooting :

Effective error handling and troubleshooting ensure the smooth operation of the PolicyGPT platform. This section describes common issues that may occur during application setup, execution, and user interactions, along with their possible solutions.

Port Conflicts

Error Description: The application may fail to start if the required port is already being used by another process.

Troubleshooting Steps:

- Check whether another process is using the application's port.
- On Windows, use:

```
netstat -ano | findstr :3000
```

Stop the Conflicting Process:

```
taskkill /PID <PID> /F
```

Change the Port: Update the **PORT** value in the **.env** file if another port needs to be used.

Database Connection Failures

Error Description: The application may fail to connect to MongoDB due to an incorrect connection string, invalid credentials, or network configuration.

Troubleshooting Steps:

- Verify the **MONGO_URI** value in the **.env** file.
 - Ensure the MongoDB username and password are correct.
 - Check whether the MongoDB database or cluster is running and accessible.
 - Verify network access and IP configuration in MongoDB Atlas.
 - Ensure the database user has sufficient permissions.
-

Dependency or Module Errors

Error Description: The application may display a **MODULE_NOT_FOUND** error when a required dependency is missing.

Troubleshooting Steps:

- Run `npm install` to install all project dependencies.
- Install a specific missing package when required:

`npm install <package-name>`

- Ensure `package.json` contains all required dependencies.
 - Restart the development server after installing the package.
-

API Request Failures

Error Description: API requests may fail because of incorrect endpoints, server configuration, network issues, or backend server errors.

Troubleshooting Steps:

- Verify that the backend server is running.
 - Check the API endpoint and request method.
 - Confirm that the frontend API URL is correctly configured.
 - Check request parameters, request body, and response errors.
 - Use Postman or browser developer tools to test API requests.
-

Authentication Issues

Error Description: Users may experience login failures, unauthorized access, or session expiration due to authentication or token-related issues.

Troubleshooting Steps:

- Verify that the correct username/email and password are being used.
 - Check whether the `JWT_SECRET` is properly configured.
 - Ensure the authentication token is included in protected API requests.
 - Clear expired tokens and log in again.
 - Verify role-based access permissions for Citizens, Government Officials, and Super Admins.
-

File Upload Failures

Error Description: Users may encounter issues while uploading documents required for scheme applications or verification.

Troubleshooting Steps:

- Check whether the required file type is supported.

- Verify file size restrictions configured on the server.
 - Ensure the `multer` dependency is installed and correctly configured.
 - Confirm that the upload directory exists and has sufficient storage.
 - Check backend logs for detailed upload errors.
-

Frontend Rendering Issues

Error Description: UI components may fail to load or display incorrectly because of import errors, dependency issues, or runtime errors.

Troubleshooting Steps:

- Check the browser developer console for errors.
 - Verify component import paths and exported component names.
 - Ensure all required frontend dependencies are installed.
 - Run `npm install` if packages are missing.
 - Restart the Vite development server after dependency changes.
-

General Troubleshooting Tips

- Check application logs for detailed error messages.
- Verify environment variables in the `.env` file.
- Restart the frontend and backend servers after configuration changes.
- Use browser developer tools and Postman for debugging.
- Run `npm install` again if dependency-related issues occur.

Maintenance and Support :

The **Maintenance and Support** section describes the procedures required to keep the PolicyGPT platform secure, reliable, and up to date. Regular maintenance helps ensure smooth application performance and provides continuous support for users.

Backup and Database Management

Regular backups should be maintained to prevent data loss and ensure system recovery when required.

Key Features:

- Regular backup of MongoDB database data.
- Secure storage of backup files.
- Monitoring database availability and performance.
- Recovery procedures in case of data loss or system failure.

Regular Updates and Security

The application should be regularly updated to maintain security and improve overall performance.

Key Features:

- Updating project dependencies and packages.
- Fixing security vulnerabilities and bugs.
- Reviewing and updating environment configurations.
- Maintaining secure JWT secrets and database credentials.

Monitoring and Error Handling

Continuous monitoring helps identify application issues and performance problems at an early stage.

Key Features:

- Monitoring frontend and backend application logs.
- Tracking server errors and API failures.
- Identifying performance and availability issues.
- Debugging application errors using logs and developer tools.

User Support

User support helps Citizens, Government Officials, and Super Admins resolve issues while using the PolicyGPT platform.

Key Features:

- Providing feedback and support functionality.
- Assisting users with login and authentication issues.
- Resolving scheme application and eligibility-related queries.
- Handling reported issues and improving the platform based on user feedback.

Version Control and Updates :

Version control is used in the PolicyGPT project to manage source code changes, support collaboration among team members, and maintain a structured development workflow. Git is used for local version control, while GitHub serves as the remote repository for storing and sharing project code.

Version Control Strategy

- The **main branch** is maintained for stable and production-ready code.
- The **dev branch** is used for integrating completed features and ongoing development work.

- Separate **feature branches** are created for specific modules or functionalities to avoid conflicts between team members.
- Changes are reviewed and merged through **Pull Requests (PRs)** before being integrated into the development branch.

Branching Workflow

The project follows a feature-based branching strategy. Each developer works on a separate branch for an assigned feature or module.

Example branch names include:

- `feat/ai-assistant-backend`
- `feat/analytics-reports-live-data`
- `feat/chat-history-sessions`
- `feat/maker-checker-rbac`
- `feat/notifications-integration`
- `feat/production-config`

This approach allows multiple team members to work simultaneously without directly affecting the stable development code.

Commit Messages

Meaningful commit messages are used to clearly describe the changes made in each update. A standard format is followed to improve readability and maintain project history.

Examples:

- `feat: Add AI assistant backend`
- `fix: Resolve notification integration issue`
- `feat: Implement role-based access control`
- `fix: Resolve database configuration issue`
- `docs: Update project documentation`

Pull Request and Code Integration

After completing a feature, the developer pushes the feature branch to the remote GitHub repository and creates a Pull Request. The changes are reviewed and tested before merging them into the `dev` branch. Once the development version is stable and tested, it can be merged into the `main` branch.

Project Updates and Maintenance

The project is regularly updated to improve functionality, fix bugs, and integrate new features. Developers synchronize their local repositories with the latest remote changes before starting new work.

Common Git commands used for project updates include:

git pull origin dev

git status

git add .

git commit -m "commit message"

git push origin branch-name

Glossary :

Policy: A government-defined set of rules, guidelines, or regulations created to address specific public needs.

Government Scheme: A government initiative designed to provide benefits, financial assistance, services, or support to eligible citizens.

Eligibility Checker: A feature that evaluates a citizen's profile and determines which government schemes they may be eligible for.

Role-Based Access Control (RBAC): A security mechanism that provides different permissions and features based on user roles such as Citizen, Government Official, and Super Admin.

Application Tracking: A feature that allows citizens to monitor the current status of their submitted scheme applications.

AI Assistant: An intelligent conversational feature that helps users search for policies and schemes and obtain relevant information.

Super Admin: A system administrator with advanced permissions to manage users, policies, schemes, and overall platform operations.

Frequently Asked Questions :

How can I check my eligibility for a government scheme?

Users can access the Eligibility Checker, provide the required profile information, and receive relevant scheme recommendations based on the eligibility criteria.

How can I apply for a government scheme?

Citizens can select an available scheme, review its details and eligibility requirements, and submit an application through the platform.

How can I track my application status?

Users can access the Application Tracking section to view whether their application is pending, approved, or rejected.

What happens after an application is submitted?

The application is forwarded to the relevant Government Official for review, who can approve or reject it based on the provided information and eligibility requirements.

What can the AI Assistant do?

The AI Assistant helps users obtain information about government policies and schemes and supports users in finding relevant information more efficiently.